SUMAN BHATTACHARYYA

Researcher in Stellar Astrophysics

E-mail link

About Me

I am an early-career observational astrophysicist whose research focuses on time-domain stellar astrophysics, combining spectroscopy, photometry, statistical analysis, and machine learning to investigate stellar variability. My research has primarily explored the evolution of circumstellar disks, binary interactions, and transient phenomena in Classical Be stars using large multi-epoch datasets from ground- and space-based observatories including ESO, TESS, Gaia, LAMOST, and BeSS.
My broader scientific interests lie in time-domain astronomy, large astronomical surveys, stellar populations, and data-intensive astrophysics. I am particularly interested in developing computational and machine-learning techniques for the analysis of survey data, with the long-term goal of advancing our understanding of stellar evolution, variable stars, and the physical processes governing transient astrophysical phenomena.

Research Interest

Time-domain astronomy, Stellar variability, Classical Be stars, Emission-line stars,Spectroscopy and Multi-epoch spectroscopy, Machine Learning, Large astronomical surveys

Contact information

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Google Scholar: scholar.google.com/citations?user=bjZY49sAAAAJhl=en

Education and Academic Degrees

2021 – 2026 : **PhD**, *The Dept. of Physics & Electronics, CHRIST (Deemed to be University)*, Bangalore.

June 2024 – May 2025 : **ESO Studentship**, *European Southern Observatory (ESO) (Vitacura, Chile)*.

2019 – 2021 : **MPhil Scholar**, *CHRIST (Deemed to be University)*, Bangalore.
CGPA (10 point scale): 9.79

2016 – 2018 : **M.Sc (Physics)**, *at University of Kalyani,*, West Bengal.
Percentage: 60.8

Research Employment

June 2021 – May 2026 : **PhD in Stellar astrophysics**, *CHRIST (Deemed to be University)*, Bangalore., Thesis Topic: BINARY INTERACTIONS AND TRANSIENT CIRCUMSTELLAR DISKS IN Be STARS: A MULTI-EPOCH SPECTROSCOPIC AND PHOTOMETRIC STUDY.
Supervisor: Dr. Blesson Mathew, Associate Professor, PhD coordinator, Dept. of Physics & Electronics.

June 2024 – May 2025 : **Studentship at European Southern Observatory-ESO (Vitacura, Santiago, Chile).**, ESO supervisors: Dr. Thomas Rivinius and Dr. Robert Klement..

June 2019 – June 2021 : **MPhil in Physics (specialization: astrophysics)**, *CHRIST (Deemed to be University), Bangalore*, Degree awarded on October 03, 2021.
Thesis Title: Investigating the evolution of emission-line stars from pre-main sequence to main sequence phase.
Supervisor: Dr. Blesson Mathew, Assistant Professor, Dept. of Physics & Electronics.

2016 – 2018 : **M.Sc (Physics)**, *at University of Kalyani,*, West Bengal.
MSc thesis on reconstruction of the Hubble Diagram using Abell catalog, under the supervision of Prof. Jyoti Prasad Saha, Professor at physics department of University of Kalyani, India.

List of Publications (Published/ Accepted)

(Most recent publications first). The format is author list (my own name in bold), publication year, journal abbreviation, volume number, issue number, first page, title, DOI.

Journal Papers

1. **Bhattacharyya, Suman**., Mathew, Blesson., Ezhikode, Savithri H., Muneer, S., G, Selvakumar., G, Maheswer., Arun, R., Anilkumar, Hema., Banerjee, Gourav., Kumar, Pramod., Kartha, Sreeja S., Paul, K. T., & Velu, C. *"Decoding the X-Ray Flare from MAXI J0709-159 Using Optical Spectroscopy and Multiepoch Photometry."* 2022, ApJL, 933, 2, L34. DOI: 10.3847/2041-8213/ac7b8a

2. **Bhattacharyya, Suman**., et al. 2026 *"A multi-epoch spectral and photometric study to understand the nature of the interesting Be star candidate HD 249179."* (Accepted in Research in Astronomy and Astrophysics (RAA)). DOI: 10.1088/1674-4527/ae983e

3. **Bhattacharyya, Suman.**, Mathew, Blesson., Banerjee, Gourav., Anusha, R., et al. "Identification of emission-line stars in transition phase from pre-main sequence to main sequence" , 2021, MNRAS, 507, 3, 3660. DOI: 10.1093/mnras/stab2385

4. **Bhattacharyya, Suman.**, Mathew, Blesson. et al. 2023,. "Insights on the Optical and Infrared Nature of MAXI J0709-159: Implications for High-Mass X-ray Binaries" vol. 93, no. 2, 2024, BSRSL, pp. 636–647. DOI:10.25518/0037-9565.11821.

5. Anilkumar Hema, et al, "A comprehensive multi-wavelength study of two X-ray emitting Be stars", Accepted in RAA. DOI: https://doi.org/10.48550/arXiv.2609.28719

6. Labadie-Bartz, J. et al 2025, "The birth of Be star disks I. From localized ejection to circularization", 2025, A&A, 699, A82. DOI: 10.1051/0004-6361/202453321

7. Ezhikode S. H., Anilkumar H., Arun R., Mathew B., Jithesh V., **Bhattacharyya S**., Nedhath S., et al., "Serendipitous detection of an intense X-ray flare in the weak-line T Tauri star KM Ori with SRG/eROSITA" 2025, PASA, 42, e009. DOI: 10.1017/pasa.2024.98

8. Akhila D., Mathew B., Nidhi S., Shridharan B., Arun R., Anilkumar H., Maheswar G., et al. "Analysis of forbidden neon emission lines in HAeBe stars using Spitzer IRS spectra ", 2025, MNRAS, 540, 3330. DOI: 10.1093/mnras/staf914

9. Edwin D., Mathew B., Shridharan B., Valsan V., Nidhi S., **Bhattacharyya S**., Kartha S. S., et al. "Spectroscopic Study of Late-type Emission-line Stars Using the Data from LAMOST DR6", 2024, RAA, 24, 125014. DOI: 10.1088/1674-4527/ad8f9b

10. Sidharth et al, "Detection of a new sample of Galactic white dwarfs in the direction of the Small Magellanic Cloud", A&A, 690, A68 (2024). DOI: 10.1051/0004-6361/202450292

11. Dewangan L. K., Maity A. K., Mayya Y. D., Bhadari N. K., **Bhattacharyya S**., Sharma S., Banerjee G., *"New*

insights in the bubble wall of NGC 3324: intertwined sub-structures and a bipolar morphology uncovered by JWST", 958 51 ApJ (2023). DOI: 10.3847/1538-4357/ad004b

12. Banerjee, Gourav., Mathew, Blesson., **Bhattacharyya, Suman.**, Devaraj, Ashish., Kartha, Sreeja S., & Joshi, Santosh. "Study of the Balmer decrements for Galactic classical Be stars using the Himalayan Chandra Telescope of India." Bulletin de la Societe Royale des Sciences de Liege, Vol-93, No. 2, pp. 504-513 (2024). DOI: 10.25518/0037-9565.11771

13. Nidhi, S., Mathew, B., Shridharan, B., **Bhattacharyya, S.**, Edwin, D., and Kartha, S. S., "Estimation of stellar parameters and mass accretion rate of classical T Tauri stars from LAMOST DR6", Journal of Astrophysics and Astronomy, vol. 44, no. 2, 2023. DOI: 10.1007/s12036-023-09965-5

14. Madhu Kashyap Jagadeesh., Mathew, Blesson., Paul, KT., Banerjee, Gourav., **Bhattacharyya, Suman.**, Anusha, R., & Kumar, Pramod., "Optical spectroscopy of classical Be stars in old open clusters". 2023, RAA, 23, 3, id.035002.DOI: 10.1088/1674-4527/acafc4

15. B. Shridharan, Blesson Mathew, **Suman Bhattacharyya**, T. Robin, R. Arun, Sreeja S Kartha, P. Manoj, S. Nidhi, G. Maheshwar, K. T. Paul, Mayank Narang, T. Himanshu, *"Emission line star catalogues post-Gaia DR3. A validation of Gaia DR3 data using the LAMOST OBA emission catalogue."*. 2022, A&A, 668, A156. DOI: 10.1051/0004-6361/202244353

16. Banerjee, Gourav., Mathew, Blesson., Paul, K. T., Subramaniam, Annapurni., Anjusha B., **Bhattacharyya, Suman**., Anusha, R., Deeja Moosa., C S Dheeraj., Aleeda Charly., & Megha Raghu. *"Study of the transient nature of classical Be stars using multi-epoch optical spectroscopy"*. 2022, JApA, 43, 102. DOI: 10.1007/s12036-022-09891-y

17. Anusha, R., Mathew, Blesson., Shridharan, B., Arun, R., Nidhi, S., Banerjee, Gourav., Kartha, Sreeja S., Paul, K. T., **Bhattacharyya, Suman.** *"Identification of new classical Ae stars in the Galaxy using LAMOST DR5"*. 2021, MNRAS, 501, 4, 5927. DOI: 10.1093/mnras/staa3964

18. Banerjee, Gourav., Mathew, Blesson., Paul, K. T., Subramaniam, Annapurni., **Bhattacharyya, Suman.**, Anusha, R,
"Optical spectroscopy of Galactic field Classical Be stars". 2021, MNRAS, 500, 3, 3926. DOI: 10.1093/mnras/staa3469

19. Bikramaditya Ghosh, Dimitris Kenourgios, Antony Francis & **Suman Bhattacharyya**. "How well the log periodic power law works in an emerging stock market?", 2021, Applied Economics Letters, 28:14, 1174-1180. DOI: 10.1080/13504851.2020.1803484

Research articles

1. **Bhattacharyya, Suman**., Mathew, Blesson., Banerjee, Gourav., Anusha, R., Paul, K. T., Kartha, & Sreeja S., *Identification of a rare class of emission-line stars in transition between pre-main sequence to Main Sequence phase*, MJS, 20, 4, 33, DOI: 10.12723/mjs.59.3

Successful Observing Proposals

▪ Proposal cycles: 2026 (HCT-2026-C1-P8 and HCT-2026-C2-P24) **Title: "Spectroscopic monitoring program to address the line profile changes of H-alpha and other metallic emission lines in classical Be stars" As Co-PI** Instrument: HFOSC , Himalayan Chandra Telescope(HCT), IIA.

▪ Proposal cycles: 2026 (VBT-2026-C1-P5) **Title: "K-HiReSS: Kavalur High-Resolution Spectroscopic Survey of bright and interesting Be stars using VBT Echelle spectrograph" As Co-PI** Instrument: Echelle spectrograph (R=27000) of the 2.3-m telescope, VBO, Kavalur, IIA.

- Proposal Accepted: Proposal cycle: P115 **Title: "Exploring the Transition Phase of Rapidly Rotating Massive Stars: A Spectroscopic Study of Herbig Ae/Be to Be Stars "** Instrument Requested: XSHOOTER (VLT UT3). PI: Suman Bhattacharyya, Co-I: Dr. Blesson Mathew, Dr. Th. Rivinius, Dr. Robert Klement.

- Proposal cycle: August, 2023 to January, 2024 (App. No. 2023B034)
Title: "Infrared line analysis of Transition Phase candidates: stars in transition from pre-main sequence to main sequence phase"
Instrument: SpeX instrument of the NASA-IRTF telescope, Mauna Kea, Hawaii Islands.
Being the PI of this project, we have received 10 hours for observation.

- Proposal cycle: February - July, 2023 (App. No. 2023A041)
Title: "Infrared line analysis of Transition phase candidates: stars in transition from pre-main sequence to main sequence"
Instrument: SpeX instrument of the NASA-IRTF telescope, Mauna Kea, Hawaii Islands.
Being the PI of this project, we have received 5.5 hours for observation of our target 13 Transition phase candidates.

- Proposal cycles: 2023 (C1, C3) **Title: "Investigating the transient nature of classical Be stars using multi–epoch optical spectroscopy"** Instrument: UAGS spectrograph (1800 lines/ mm grating) of the 1-m telescope, VBO, Kavalur, IIA. Being a Co-PI of this project, I have till now received 58 nights for observation of our target 23 Be stars.

- Proposal cycle: 2023 (C1)
Title: "Photometric study of the BeXRB system HD 54786"
Instrument: UKATC instrument of the 1.3-m J.C. Bhattacharyya Telescope, Kavalur.
Being a PI of this project, we have received 8 nights for observation of the target star.

- Proposal cycles: 2023 (C1, C2)
Title: "Investigating the transient nature of classical Be stars using multi- epoch optical spectroscopy"
Instrument: UAGS spectrograph (1800 lines/ mm grating) of the 1.02-m telescope, VBO, Kavalur, IIA.
Being the Co-PI of this project, I have received 20 nights for observation of our target 16 Be stars.

- Proposal cycles: 2022 (C1, C3)
Title- "Spectroscopic survey of emission-line stars in transition phase from pre-main sequence to main sequence"
Instrument: Eschelle spectrograph of 2.34-m Vainu Bappu Telescope, Kavalur.
Being a PI of this project, we have received 11 + 8 nights for observation of our target 35 emission-line stars.

- Proposal cycles: 2022 (C3) **Title- "Spectroscopic monitoring of the line variability in various mass ranges of Herbig Ae/Be stars"**
Instrument: Eschelle spectrograph of 2.34-m Vainu Bappu Telescope, Kavalur.
Being a Co-PI of this project, we have received 3 + 3 nights for observation of our target 9 HAeBe stars.

Computational Skills

Programming and scientific computing:	**Python; for astronomical data analysis, numerical computation and visualisation; Fortran (Medium level); R (Medium level)**.
Machine learning and deep learning:	**Experience implementing convolutional neural networks (CNNs) and ResNet-based approaches using Keras/TensorFlow for astronomical classification and data-driven analysis.**.

Time-series Analysis:	**Lomb-scargle periodograms (Period04, Lightkurve),variability analysis, light-curve processing, detrending, interpolation, binning, statistical analysis**.
Large-survey data analysis:	**Processing and cross-matching heterogeneous astronomical catalogues and observations from LAMOST, Gaia, TESS, BeSS, 2MASS and WISE using Python-based workflows.**.
Representation learning (Medium level) and visualisation:	**Developing an interest in learned spectral representations, dimensionality reduction (UMAP), clustering, anomaly detection and uncertainty-aware classification for large spectroscopic datasets.**.
AI-assisted development:	**Currently exploring AI-assisted programming tools, including Claude Code, to support code development and debugging, Fine tuning accessible LLM models (GPU based) utilizing Ollama interface (Medium level)**.

Technical & Observational Skills

Photometry:	**TESS light curves, Gaia photometry, AAVSO, 2MASS, WISE, long-baseline photometric monitoring**.
Spectroscopy:	**Multi-epoch optical spectroscopy analysis, emission-line diagnostics, spectral fitting, SED, Spectral Homogenization**.
Surveys & Archives:	**TESS, Gaia DR3, LAMOST, BeSS, AAVSO, 2MASS, ALLWISE, IPHAS/VPHAS+**.
Observational Facilities:	**ESO/VLT, HCT, VBT, VBO, IRTF**.

Honours/Awards/Grants received

- Served as Scientific Assistant in P116 OPC Expert Panel Meeting of ESO observing proposals (2025).
- European Southern Observatory Studentship in Chile, 06/2024 - 5/2025
- Received the Research Excellence Award at the annual Scholar Meet (2023) by CHRIST University
- SEED Money project grant Dec 2022 to March 2023, CHRIST University, Bangalore, India
- MRP (Major Research Project) 2022-2023 funding, CHRIST University, Bangalore, India

Conferences

1. Poster Presentation on long term disk study of classical Be stars through data-driven approach, in "Stellar Variability: Taking the Pulse of the Universe", November 24 - 28, 2025, IUCAA, Pune, India

2. Poster Presentation (Title: Tracing the Rhythms of Circumstellar Disks: Variability and Duty Cycles in Classical Be Stars) in "The role of accretion and ejection variability in the evolution of young stars and their disks" (RAVEYSO), May 19-22, 2025. Organized by ESO, Garching at ESO headquarters in Garching, Munich.

3. Contributory talk (Title: "Transforming Be star research through machine learning and big data: A new approach in emission-line star studies") in Astroinformatics 2024, December 9-13, 2024 - Puerto Natales, Patagonia, Chile. Organized by University of Concepción, YEMS collaboration, Instituto Milenio De Astrofisica, H-ITS (Germany).

4. Selected for a talk (Virtual) in "Hot Stars 2024" Life With Circumstellar Matter", October 14-19, 2024, at Almaty, Kazakstan. Title: "Exploring Be Star Duty Cycles: A Data-Driven Study of Hα Spectral Variations".

5. Participation in ESOGPT24 online conference, 23 - 27 September 2024. Sponsored by ESO Garching.

6. Contributory talk (Title: "Unraveling the recent X-Ray flare from MAXI J0709–159 using Indian optical telescope facilities") in the 5th National Conference on 'REcent Trends in the study of Compact Objects: Theory and Observation (RETCO-V)' organized by Indian Intitute of Astrophysics, India at the Kodaikanal Solar Observatory, Tamil Nadu, India during April 03 – 05, 2023.

7. Contributory talk (Title: "Study of the recent X-Ray flare from MAXI J0709–159 using Indian optical telescope facilities") in the '3rd Belgo-Indian Network for Astronomy & Astrophysics' Workshop (BINA 2023) hosted at ARIES, Nainital during March 22 – 24, 2023.

8. Contributory talk (Title: "Identification of a special class of emission-line stars between PMS and MS phase") in the 'Regional Astronomy Meeting' (RAM 2023) sponsored by IUCAA at Rajagiri School of Engineering and Technology (RSET), Kerala, India during February 17 – 19, 2023.

9. Poster (Title: Unraveling the X-ray flare from MAXI J0709–159 using optical photometry and spectroscopy) in 'Young Astronomers' Meet 2022' (YAM-2022) hosted by ARIES, Nainital during 9th-13th November, 2022 via online mode.

10. Poster (Title: A WISE mid infrared analysis of classical Be stars) in 'XXXXth Astronomical Society of India Meeting' (ASI-2022) hosted by IIT Roorkie and ARIES, Nainital during March 25 – 29, 2022 via online mode.

11. Flash Poster (Title: Identification of a rare class of emission-line stars between PMS and MS phase) in '21st National Space Science Symposium' (NSSS 2022) hosted by IISER-Kolkata during January 31 – February 04, 2022 via online mode.

12. Paper (Title: Emission line stars in the transition phase from pre-main sequence to main sequence) in 'Regional Astronomy Meeting VII' (RAM-VII) hosted by the sponsorship of IUCAA from September 08 – 10, 2021 via online mode.

13. Poster (Title: Transition candidates from the pre-main sequence phase to the main sequence phase) in 'Regional Astronomy Meeting VI' (RAM-VI) organised jointly by the Departments of Physics at Mar Thoma College, Chungathara and Providence Women's College, Calicut in collaboration with IUCAA from July 09 – 10, 2020 via online mode.

14. Paper (Title: Reconstruction of Hubble constant based on galaxy clusters using SDSS database.) in 'International E-Conference on Recent Advances in Physics: A Promise to Society' (IC-RAPPS 2020) hosted by Govt. PG College, Bajpur, Uttarakhand during June 24 – 25, 2020 via online mode.

15. Poster (Title: Transition candidates from pre-main sequence to main sequence phases) in 'XXXVIIIth Astronomical Society of India Meeting' (ASI-2020) hosted by IISER Tirupati during March 13 – 17, 2020.

Mentoring

I have co-supervised the following MSc (Physics) students for their Master's dissertation.

- Deepika Patel, CHRIST (Deemed to be University), Ongoing from December 2025.

Topic: emission line stars.

- Gaganashree S, CHRIST (Deemed to be University), Ongoing from April, 2023.

Topic: Study of Runaway stars.

- Sidharth A V and Shriranjani Sowmianarayanan, CHRIST (Deemed to be University), submitted in 2023.
Topic: V/R variation study of Classical Be stars using LAMOST spectral survey.

- Aleeda Charley and Megha Raghu, CHRIST (Deemed to be University), submitted in 2021.
Title: Studying the Hα variability of Galactic Classical Be stars.

Media Coverage

1. Invited by 'R Plus news' (one of the leading news channel in Bengali in India) to take part in a live discussion program on India's chandrayaan 3 Mission on August 23, 2023, the day India's landing near to the south pole of the Moon successfully.

2. Interviewed by 'SHAASTRA' magazine (one of the leading science magazine in India by IIT Madras) to talk about the identifying of a new category of stars in transition phase based on my Bhattacharyya et al. 2021 paper. The coverage was published in their Volume 01 Edition 03 Sep-Oct 2021. https://shaastramag.iitm.ac.in/

Participation in selected workshops/ schools/ training camps

1. Participated in *AstroAI Workshop on June 15-18, 2026 (online)* at Center for Astrophysics, Harvard & Smithsonian (CfA), 60 Garden St., Cambridge, MA 02138.

2. Participated in *FAR-SEE 2026 (Frontiers of AI Research in Space, Earth System and Engineering Sciences)*, the AI-ML Vikram Discussions organised by PRL, Ahmedabad.

3. 8th Indo-French Astronomy School (IFAS8), Spectroscopy 3D. IFAS8 co-organized by Lyon (LIO, CRAL CNRS UMR5574, UCBL1) and IUCAA in Pune.

4. Workshop on 'HPC Workshop on Radio Astronomy' conducted as a part of the 40th ASI meeting (ASI-2022), on March 25, 2022.

5. Workshop on 'Ultraviolet Space Astronomy : UVIT and Beyond' conducted as a part of the XXXIXth ASI meeting (ASI-2021), on February 18, 2021.

6. Winter School in Astronomy, Astrophysics and Cosmology (WSAAC 2019) during January 07 – 13, 2019 sponsored by IUCAA held at Guahati University.

7. Workshop on 'Observing Universe with ASTROSAT' during September 03 – 05, 2018 at the Manipal Centre for Natural Sciences, MAHE, Manipal. Arranged by IUCAA

Participation as Resource person

- Supported and mentored students in Astronomy Astrophysics Introductory Lectures Lab (AAILL-26), December 19, 2025 – January 9, 2026 at Vainu Bappu Observatory, Indian Institute of Astrophysics, Kavalur, India.

- Co-ordinator of Young Astronomers' Meet (YAM) 2024 conference, Mar 6 – 9, 2024 at CHRIST (Deemed to be University), Bangalore, India.

- Two days Hands-on Session on Astronomical Data Processing.
Organized by Research & Postgraduate Department of Physics, Bharat Mata College, Thrikkakara on 23 – 24 February 2023.

- NSAASSR 2023 Conference in collaboration with IUCAA, Pune, India on 13 – 15th February 2023.

Public Outreach Involvement

1. Resource person during ESO Open day at Santiago, Chile. March 21, 2025.

2. Delivered more than 70+ popular astronomy talks at various schools, colleges, associations, organizations for popularizing astronomy at different parts of India since 2017.

3. Participated as a resource person in 3 astronomy camps and workshops organized by schools, colleges/University (Kalyani University), amateur astronomy associations (Sky Watchers Association, Kolkata Astronomy centre, Kolkata Institute of Fundamental Science and Astronomy) for popularizing astronomy across the nation since 2017.

Interests

- Cycling is a hobby of mine and one of the favorite pastime. I enjoy exploring new routes and pushing myself to go further and faster with each ride. It serves as a great stress-reliever and allows me to immerse myself in the beauty of nature.

- Painting and photography are two of my favorite hobbies that allow me to express my creativity in different ways. As photography enthusiast, astrophotography allows me to explore the beauty of the night sky and appreciate the wonders of the universe in a whole new way.

Referees

Referee 1 **Dr. Blesson Mathew,**
Associate Professor, Department of Physics & Electronics
CHRIST (Deemed to be University)
✉ blesson.mathew@christuniversity.in

Referee 2 **Dr. Maheswar Gopinathan**
Professor
Indian Institute for Astrophysics, Bangalore
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Referee 3 **Dr. Sreeja S Kartha,**
Assistant Professor, Department of Physics & Electronics
CHRIST (Deemed to be University)
✉ sreeja.kartha@christuniversity.in

Referee 4 **Dr. Thomas Rivinius,**
Professor
European Southern Observatory, chile
✉ triviniu@eso.org

Referee 5 **Dr. Robert Klement,**
Fellow
European Southern Observatory, chile
✉ Robert.Klement@eso.org

Declaration

I hereby declare that the above mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars.